

## Reviewed Preprint

v1 • July 24, 2024

Not revised

## Reviewed Preprint

v2 • March 12, 2026

Revised by authors

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### Competing interests:

No competing interests declared

### Funding:

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### Reviewing editor:

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# Human adherent cortical organoids in a multiwell format

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## eLife Assessment

This paper describes an **important** advance in a 2D in vitro neural culture system to generate mature, functional, diverse, and geometrically consistent cultures, in a 384-well format with defined dimensions and the absence of the necrotic core, which persists for up to 300 days. The well-based format and conserved geometry make it a promising tool for arrayed screening studies. The evidence is **compelling** and provides a method for generating consistent 3D cortical layer-like organization.

<https://doi.org/10.7554/eLife.98340.2.sa2>

## Abstract

In the growing diversity of human iPSC-derived models of brain development, we present here a novel method that exhibits 3D cortical layer formation in a highly reproducible topography of minimal dimensions. The resulting adherent cortical organoids develop by self-organization after seeding frontal cortex patterned iPSC-derived neural progenitor cells in 384-well plates during eight weeks of differentiation. The organoids have stereotypical dimensions of 3 x 3 x 0.2 mm, contain multiple subtypes of neurons, astrocytes and oligodendrocyte lineage cells, and are amenable to extended culture for at least 10 months. Longitudinal imaging revealed morphologically mature dendritic spines, axonal myelination, and robust neuronal activity. Moreover, adherent cortical organoids compare favorably to existing free-floating brain organoid models on the basis of robust reproducibility in obtaining topographically-standardized radial cortical structures and circumventing internal necrosis. Adherent human cortical organoids hold considerable potential for high-throughput drug discovery applications, neurotoxicological screening, and mechanistic pathophysiological studies of brain disorders.

## Introduction

Human embryonic or induced pluripotent stem cell (hiPSC) derived models have yielded considerable success in elucidating neurodevelopmental biology in health and disease (Levy and Pasca, 2025). The approaches have been varied and complementary, including single cell hiPSC-derived models grown in a monolayer (Sarkar et al., 2018; Shan et al., 2024; Zhang et al., 2013), multiple neural cell types in 2D neural networks (Astick and Vanderhaeghen, 2018; Bardy et al., 2015; Gunhanlar et al., 2018; Shi et al., 2012), 3D free-floating regionalized neural organoids (Pasca et al., 2015; Qian et al., 2016a; Ullah et al., 2025; Xiang et al., 2019; Zhang et al., 2023), and hiPSC-derived free-floating unguided neural organoids and assembloids (Gomes et al., 2020; Kim et al., 2025; Lancaster et al., 2013; Pasca et al., 2025; Pellegrini et al., 2020; Renner et al., 2017; Sawada et al., 2020). Although models are continuing to improve, increasing cellular and topographical complexity has appeared to come at

the cost of inter-organoid variability (Eichmüller and Knoblich, 2022 [↗](#); Kelava and Lancaster, 2016 [↗](#)). Therefore, a major current technical challenge is to identify hiPSC-derived models that recapitulate higher-order neural complexity with reduced heterogeneity.

Existing 3D models suffer from considerable inter-organoid variability due to the complex and heterogeneous nature of the free-floating structures (Cederquist et al., 2019 [↗](#); He et al., 2024 [↗](#); Lancaster et al., 2013 [↗](#), 2017 [↗](#); Paşca et al., 2015 [↗](#); Renner et al., 2017 [↗](#); Velasco et al., 2019 [↗](#); Yoon et al., 2019 [↗](#)). A further challenge, in particular with free-floating organoids, is the resulting necrotic core that frequently occurs when tissue volumes exceed the limits of oxygen and nutrient diffusion beyond a radius of ~300–400µm (Lancaster et al., 2017 [↗](#); Qian et al., 2016b [↗](#); Uzquiano et al., 2022 [↗](#)). Although recent progress has been made with slicing organoids prior to the emergence of necrosis, followed by organotypic air-liquid interface culture (Giandomenico et al., 2019 [↗](#); Qian et al., 2020 [↗](#)), even sliced organoids have to be repeatedly re-cut to prevent necrosis (Qian et al., 2020 [↗](#)), which is both laborious and risks introducing another potential source of variability. The generation of vascularized organoids is currently a major focus in the field as a potential solution to reducing the necrotic core and some of the inherent cellular stress that is observed in cortical organoids. Moreover, xenotransplantation of organoids and hiPSC-derived neural cells into rodent brains also holds considerable promise as a model system to investigate circuit integration *in vivo* (Bhaduri et al., 2020 [↗](#); Fan et al., 2022 [↗](#); Paşca et al., 2025 [↗](#); Schafer et al., 2023 [↗](#); Vermaercke et al., 2024 [↗](#)), but relies upon the use of animal models that limits advantages of a fully human cellular system. Increasing diffusion in organoids through microfluidics or vascularization (Kistemaker et al., 2025 [↗](#); Matsui et al., 2021 [↗](#)) would be beneficial. Additionally, further investigation into the mutually supportive interactions between neural and vascular cells in a fully *in vitro* system composed exclusively of human cells is warranted (Crouch et al., 2022 [↗](#); Mansour et al., 2018 [↗](#); Wang et al., 2023 [↗](#)).

Here we propose a simplified approach to generating long-term hiPSC-derived adherent cortical organoids (ACO) with reproducible dimensions below metabolic diffusion limits in a readily screenable 384-well format. ACOs can be maintained in long-term culture and contain neurons with dendritic spines and robust activity, as well as several classes of glial cells including oligodendrocyte precursor cells (OPCs), myelinating oligodendrocytes, and morphologically distinct subtypes of astrocytes.

## Results

### Self-organized topography of iPSC-derived adherent cortical organoids

ACO reproducibly self-organize into layered radial structures in 384-well plates within 8 weeks of seeding with hiPSC-derived purified forebrain-patterned neural progenitor cells (NPCs). Three different hiPSC source cell lines were used in this study, including commercially available NPCs, and NPCs generated using a modified version of our previously described protocol (Gunhanlar et al., 2018 [↗](#)) (Figure 1A [↗](#)). NPCs were capable of neural rosette formation and expressed SOX2, Nestin, and the frontal cortical NPC-marker FOXG1 (Figure 1B–D [↗](#), Figure S1A). The initial 4 weeks after seeding the NPCs in 384-well plates in Neural Differentiation Medium (ND) were characterized by proliferative expansion of NPCs and the emergence of early neural differentiation markers (Figure 1E [↗](#), Figure S2). Between 4- and 8-weeks post-seeding, neurons and glial cells emerged with a consistent spatial organization (Figure 1E, F [↗](#), Figure S1B, Figure S2), in which the central region was densely packed with cell bodies while the periphery contained circumferentially and radially organized processes originating from cells in the centre. Typically, ACOs with a single structure were observed in ~80% of the wells seeded with NPCs after 60 days of differentiation. Structural integrity remained quite stable and exhibited minimal decline over time with slight variation per cell line and per batch to an average of ~65% of wells with a single structure at day 234 and ~50% at day 310. (Figure S1C). Organoid topography was highly dependent on the proliferation rate of NPCs, which can differ substantially across differentiation batches and hiPSC clones. For each NPC line, optimal seeding density was estimated based on the

proliferation rate of that NPC line. Multiple densities were seeded around the estimated optimal density, and after 6-8 weeks, the NPC density at which ACO generation were most reliably generated was determined empirically using a combination of visual inspection, Live/Dead staining and immunocytochemistry. The proliferation rate and seeding density of NPCs required for generating topographically reliable ACOs were correlated ( $r = 0.67$ ) and provides guidance for empirically determining the optimal NPC seeding density for a given line (Figure S1D).

## Cell type distribution and layer formation

Tau+/MAP2- axons exhibited long extensions in a circular pattern, while MAP2+ dendrites exhibited orthogonally oriented radial outgrowth (Figure 2A, B [Figure S3](#)). The spatial organisation that evolved over the first 8 weeks after seeding was paralleled by a shift in cell type distribution.

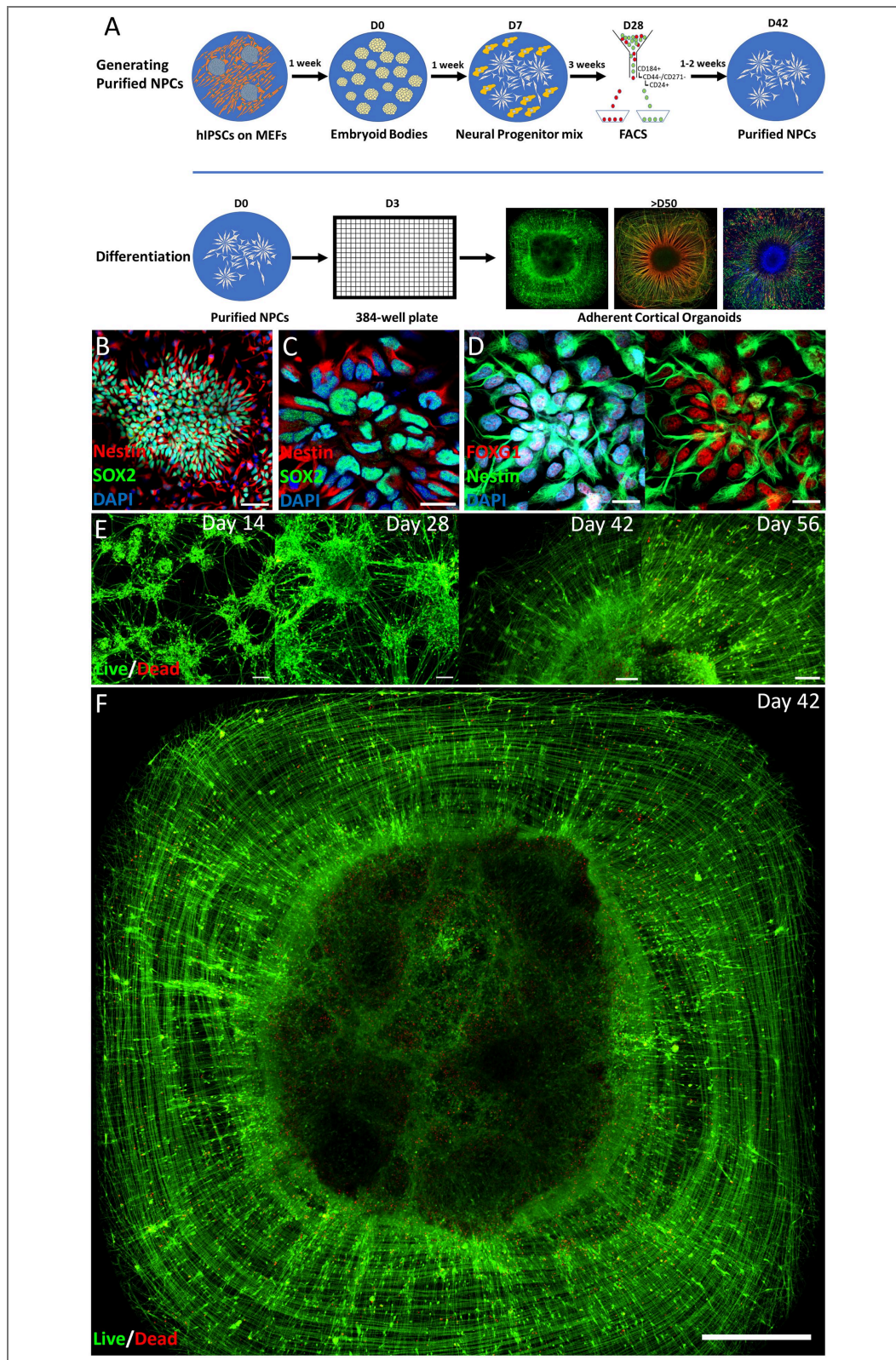
Overall, a reduction in progenitor markers (SOX2 day 14: 58.4%, day 56: 18.8%  $P = <0.001$ , PAX6 day 14: 34.5%, day 56: 8.0%  $P = 0.20$ ) and a significant increase in neuronal cortical layer markers (CTIP2 day 14: 0.5%, day 56: 14.0%  $P = <0.001$ , CUX1 day 14: 1.3%, day 56: 24.4%  $P = <0.001$ ) were observed (Figure S2). However, a SOX2 positive layer remained present in the centre of the ACOs at day 63 (Figure S4C). Cortical layer markers exhibited an inside-out pattern of development in which expression of the deep layer excitatory neuronal marker CTIP2 emerged before the upper layer marker CUX1 (Figure S2, [Figure 2C-E](#) [Figure S3](#)). Six to eight weeks following seeding, a self-organised rudimentary segregation of deep- and upper layer neurons emerged as shown by the macroscopic separation of deep- and upper layer neurons, although some neurons were spatially intermixed and some neurons were double-positive for CTIP2 and CUX1. In the upper layer neuronal region, segregation was observed between CUX1 and CUX2 positive cells ([Figure 2E](#) [Figure S3](#)) as CUX2 is typically expressed over a wider range of upper cortical layers than CUX1 and also marks intermediate progenitors (Molyneaux et al., 2007 [Figure S3](#)) ([Figure 2C, E](#) [Figure S4](#)). Analogous to the broad distribution of cortical cell subclasses, the majority of the neurons were glutamatergic, while GAD67+ interneurons were also present ([Figure 2F](#) [Figure S3](#)), constituting ~10% of the NeuN+ neuronal population consistently for all three source hiPSC lines ([Figure 2G](#) [Figure S3](#)). Over a developmental time course GAD67/NeuN remained relatively stable, although it showed a slight increase at differentiation day 114 ([Figure 2H](#) [Figure S3](#)).

## Adherent cortical organoids contain multiple glial cell types

Within 8 weeks of seeding, a population of GFAP+/S100β+ astrocytes emerged. Many astrocytes had their soma located in the central region with process outgrowth radially ([Figure 2I](#) [Figure S5](#)), while other astrocytes exhibited subtype-specific morphologies including fibrous astrocytes ([Figure 2J](#) [Figure S5](#)), protoplasmic-like astrocytes ([Figure 2K](#) [Figure S5](#)) and interlaminar astrocytes ([Figure 2L](#) [Figure S5](#)). GFAP/PAX6 double-positive radial glia were present at the outskirts of the densely populated centre of the organoid ([Figure 2M](#) [Figure S5](#)).

Similar to free-floating organoids, ACOs survived for longer periods compared to monolayer neural cultures grown on larger surfaces. This longevity allows for the development of cell types not usually seen in a monolayer culture that can typically be cultured up to a maximum of 2 to 3 months. By 6 weeks after seeding NPCs, we observed the emergence of oligodendrocyte precursor cells (OPC), as shown by the expression of NG2 ([Figure 3A](#) [Figure S6](#)) and the NG2+ cell population remained present when the first oligodendrocytes emerged ([Figure 3B/C](#) [Figure S6](#)). Staining for Myelin Basic Protein (MBP) revealed the emergence of MBP+ oligodendrocytes around 4 months after NPC seeding when the ACOs were continuously grown in the presence of T3 (2ng/ml) ([Figure 3D](#) [Figure S6](#)). At 5 months, the oligodendrocytes showed increasingly mature morphologies ([Figure 3E-I](#) [Figure S6](#)) and exhibited MBP co-localization along NF200+ axons ([Figure 3F/I](#) [Figure S6](#)).

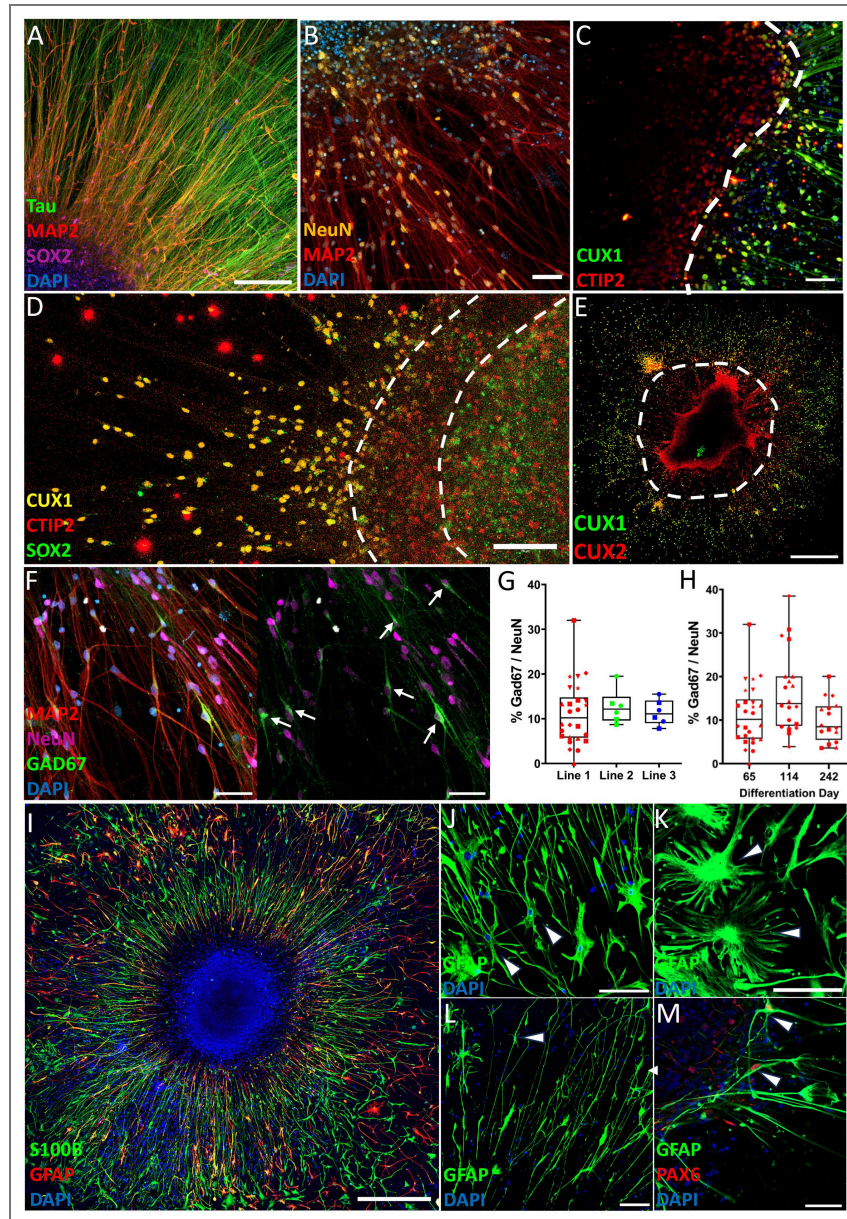




**Figure 1. Adherent cortical organoid model.**

**A** Schematic representation of the differentiation protocol. **B/C/D** Representative pictures of NPCs from three different iPSC lines with markers SOX2, Nestin and FOXG1 (scale bar B, 50  $\mu$ m; C/D, 20  $\mu$ m). **E** Live/Dead stain of a representative time course showing self-organization during differentiation, starting with radial organization between day 28 and day 42, seeding density 1500 NPCs per well; scale bars left to right 100  $\mu$ m, 150  $\mu$ m, 100  $\mu$ m, 100  $\mu$ m). **F** Full well showing radial organization at day 42 in culture where only a few dead cells are visible in red in the dense centre of the structure, seeding density 1500 NPCs per well (scale bar, 500  $\mu$ m).

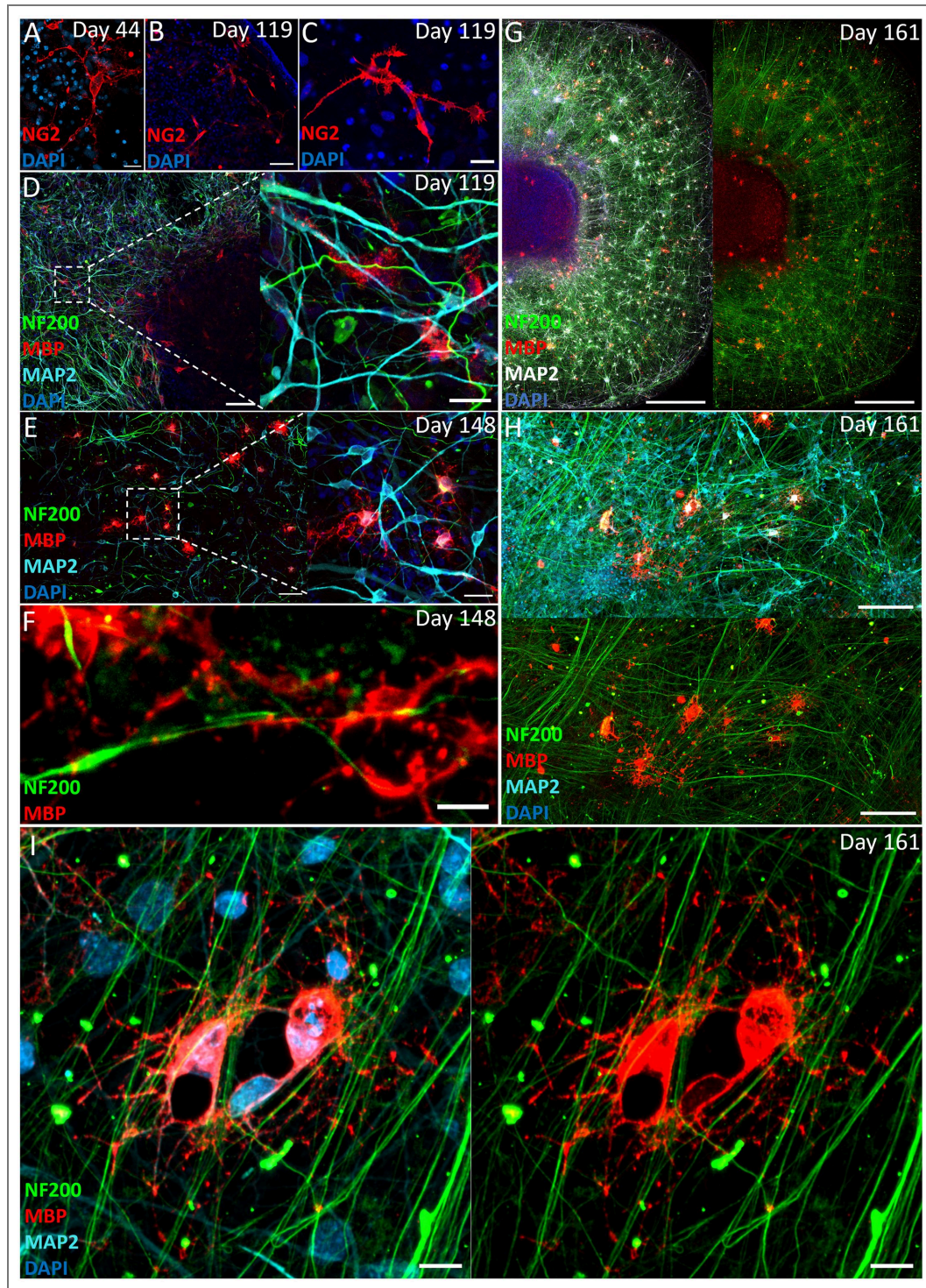




**Figure 2. Adherent cortical organoids show an organized network of neuronal and astrocyte subtypes.**

**A** MAP2+ somas and dendrites alongside Tau+/MAP2- axons show segregation of dendritic and axonal compartments, with SOX2+ progenitors concentrated in the center of the organoid (Day 75, 200µm). **B** MAP2+ and NeuN+ cells indicate mature neurons (Day 72, 50 µm). **C** Deep-layer cortical marker CTIP2 and upper layer marker CUX1 show rudimentary segregation of cortical layers in expected inside-out pattern (Day 64, 50 µm). **D** Rudimentary separation of SOX2+ NPCs, CTIP2+ deep layer neurons and CUX1+ upper layer neurons (Day 67, scale bar 100 µm) **E** Full well overview of the CUX1 and CUX2 positive regions. **F** GAD67+ interneuron (white arrows) proportion of NeuN+ neurons (Day 67, 50 µm). **G** Quantification of GAD67+ proportion of NeuN+ neurons. Every point is the percentage of GAD67+ interneurons out of all NeuN+ neurons in a spatially randomized selected image with on average 130.6 (± 16.1) NeuN+ nuclei per frame. (Line 1: N= 26 images of 6 different cortical organoids at day 65-67; Line 2/3: 6 images of 2 different organoids at day 65). **H** Quantification of GAD67+ proportion of NeuN+ neurons of cell line 1, at 3 different time points. The number of GAD67+ interneurons over the total NeuN+ population remains relatively stable although at day 114 there was a modest increase which disappeared at day 242. Data was collected from 4-6 organoids per time point. G/H The different symbols reflect from which organoid the data point was collected. **I** Astrocyte markers GFAP and S100β show the general radial pattern of astrocyte outgrowth (Day 66, 500 µm). **J/K/L** GFAP staining reveals the morphologies of different astrocyte subtypes (white arrows), including fibrous astrocytes (J, Day 65, 100 µm), protoplasmic astrocytes (K, Day 65, 50 µm) and interlaminar astrocytes (L, Day 65, 100 µm). **M** Co-localization of astrocyte marker GFAP and PAX6 marks radial glia (white arrows) (Day 65, 50 µm).





**Figure 3. Adherent cortical organoids form oligodendrocyte lineage cells.**

**A** Adherent cortical organoids show OPCs as early as 44 days, indicated by OPC marker NG2 (Day 44, 20  $\mu$ m). **B/C** The NG2+ OPCs are still present in the cortical organoids after 4 months (Day 119, B 50  $\mu$ m, C 20  $\mu$ m). **D** Young oligodendrocytes start to emerge after 4 months indicated by rudimentary MBP staining (Day 119 100  $\mu$ m and 20  $\mu$ m). **E/F** After 5 months, MBP-positive oligodendrocytes show more mature morphology and initial wrapping of NF200+ axons (Day 148; E, 50  $\mu$ m and 20  $\mu$ m, F 5  $\mu$ m). **G/H/I** Oligodendrocyte distribution at day 161 where the MBP+ oligodendrocytes sit between axons bundles and co-localize with NF200+ axons (G/H/I Day 161, G 500  $\mu$ m, H 100 $\mu$ m, I 10  $\mu$ m).

## Adherent cortical organoids show synaptic connectivity and functional activity

Neurons within ACOs exhibited robust synaptogenesis (Figure 4A-E). Sparse labelling of excitatory neurons with AAV9.CamKII.eGFP revealed the presence of Synapsin-positive (Syn+) mushroom-shaped dendritic spines (Figure 4E). 51.5% of the Syn+ presynaptic boutons co-localized with the excitatory post-synaptic density marker HOMER1 (Figure 4F, I), with a density of 10.9 Synapsin/HOMER1 double positive synapses per  $100\mu\text{m}^2$  MAP2 (Figure 4H). Moreover, 21.1% of Syn+ presynaptic boutons co-localized with the inhibitory post-synaptic density marker Gephyrin (Figure 4G, H, I), with a density of 5.0 Synapsin/Gephyrin double positive synapses per  $100\mu\text{m}^2$  MAP2. At differentiation day 114, we observed a ~2:1 ratio of excitatory to inhibitory synaptic contacts (Figure 4H).

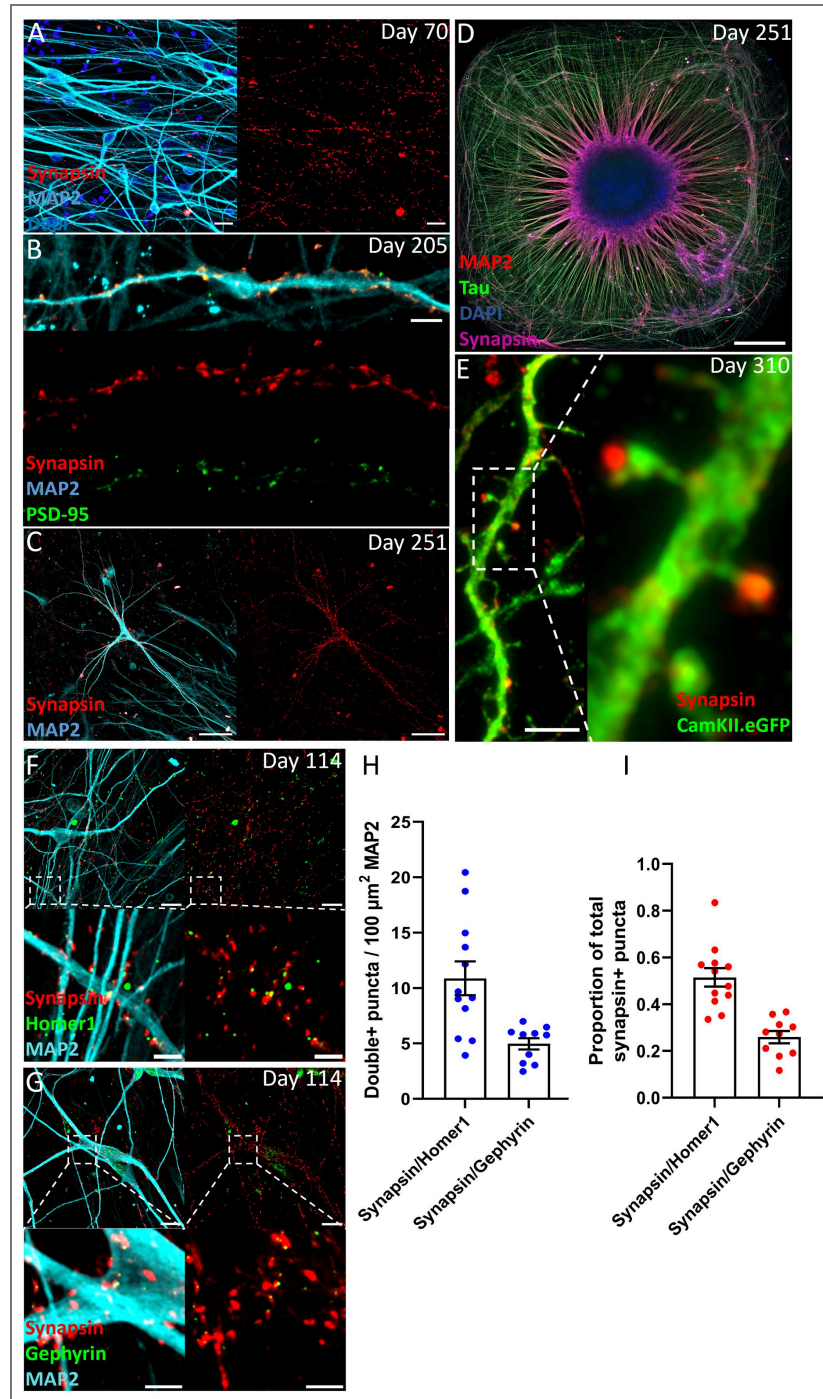
To assess the functional activity of the ACOs, we used the genetically encoded calcium indicator GCaMP6s under the control of the human Synapsin promoter (Figure 5A). Calcium imaging revealed robust synchronous network-level bursting (NB) ( $1.4 \pm 0.07$  NB/min) at day 61 which remained stable until at least day 100 ( $1.6 \pm 0.12$  NB/min), in which all recorded neurons participated to varying degrees (Figure 5B, D). In addition, substantial desynchronized activity was also observed outside of network-level bursting while the total activity remained stable over time (day 61,  $3.9 \pm 0.5$  events/min; day 100,  $3.9 \pm 0.6$  events/min) (Figure 5B-E, Video 1).

## Discussion

The study of early human brain development and related diseases has long been hampered by the inherent complexity of the human brain and the inaccessibility of living brain tissue at cellular resolution. Technological advances in induced pluripotent stem cell technology have now facilitated the opportunity to obtain living human neurons derived from specific individuals.

Here we describe a platform to model early human frontal cortical development with high reproducibility and simplified organization. While 3D free floating organoids and sliced organoids recapitulate layered cortex formation, they are subject to variation in the relative contribution of cortical tissue within the organoid, forming multiple cortical patches along the edges and complicating structured analysis (Eichmüller and Knoblich, 2022; Giandomenico et al., 2019; Quadrato et al., 2017). Moreover, 3D free floating organoids suffer from necrosis in the core of the organoid due to lack of oxygen and nutrient diffusion. Recently, other protocols have been published starting from rosette formation with a focus on early neurodevelopment and leading to single structure free floating cortical organoids (Pagliaro et al., 2023; Tidball et al., 2023). Our platform predefines a rosette-forming iPSC-derived cortical NPC population that self-organizes into adherent singular radial structures in a standard 384-well format. Other examples have highlighted the benefits of a multi-well format or systematic individual structure formation (Knight et al., 2018; Medda et al., 2016). Our platform now integrates these features to yield individual adherent layered cortical structures with synaptic connectivity and neuronal activity, as well as the major neural progenitor-derived glial cell types including OPCs, myelinating oligodendrocytes, and astrocytes. The reproducible format of adherent organoids in a 384-well format has the distinct advantage of being able to image entire organoids without slicing or clearing, including under live cell conditions. We confirmed the self-organizing potential and reproducibility of adherent cortical organoids across multiple hiPSC lines and using different sources of NPCs, controlling the seeding density for the proliferation rate of the specific NPC batch. Seeding NPCs with frontal cortical identity in the defined geometry of a 384-well plate enabled the development of long-term functional neural networks in a complex radial structure resembling early human cortical development. Future studies aimed at single-cell RNA sequencing and advanced image-based analysis solutions in this platform will be important to further benchmark adherent organoids with topographical, typological and temporal hierarchies in the developing human cortex (Bhaduri et al., 2021; Nowakowski et al., 2017, 2016; Uzquiano et al., 2022). Functional ACOs in a multi-well format have the potential to be leveraged for high-throughput

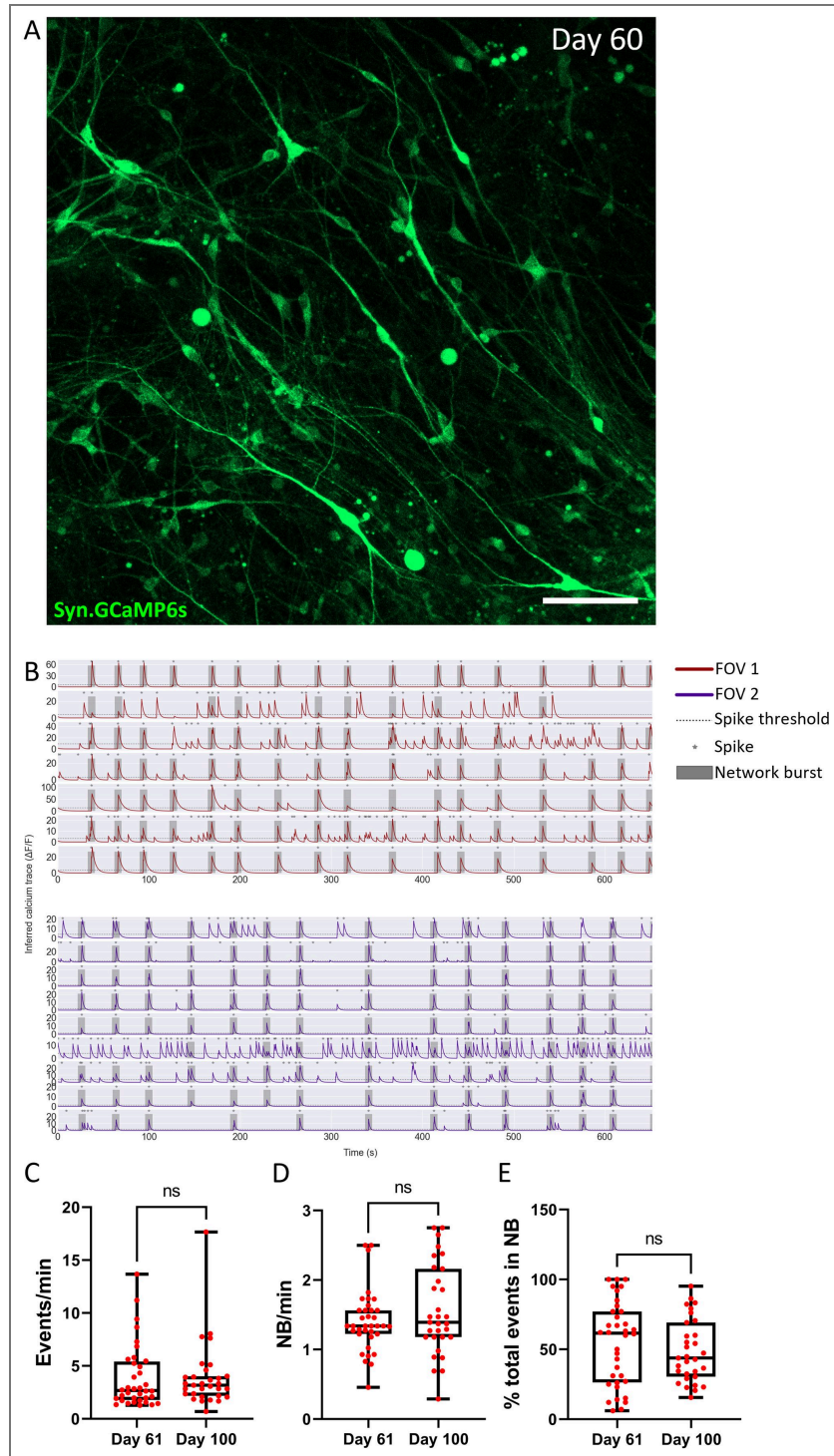




**Figure 4. Adherent cortical organoids show a neural network with excitatory and inhibitory synapses.**

**A** Synapsin staining shows synapse formation along MAP2+ dendrites (Day 70, 20  $\mu\text{m}$ ). **B** Co-localization of pre-synaptic marker Synapsin and post-synaptic marker PSD-95 (Day 205, 20  $\mu\text{m}$ ). **C** MAP2+ dendrites and soma are decorated with Synapsin+ synapses (Day 251, 100  $\mu\text{m}$ ). **D** Overview of entire well, showing alignment of Synapsin staining with MAP2 (Day 251, 500  $\mu\text{m}$ ). **E** Sparse labeling of neurons with AAV9.CamKII.eGFP allows detailed imaging of glutamatergic dendritic spines with mature mushroom morphology, contacting pre-synaptic Synapsin puncta (Day 310, 5  $\mu\text{m}$ ). **F** Excitatory synapses marked by overlapping Synapsin+ and HOMER1+ puncta (scale bar: top 20  $\mu\text{m}$ , bottom 5  $\mu\text{m}$ ). **G** Inhibitory synapses marked by overlapping Synapsin+ and Gephyrin+ puncta (scale bar: top 20  $\mu\text{m}$ , bottom 5  $\mu\text{m}$ ). **H** Density of excitatory and inhibitory synapses per 100  $\mu\text{m}^2$ . There are about twice as much excitatory synapses (10.9 / 100  $\mu\text{m}^2$ ) as inhibitory synapses (5.0 / 100  $\mu\text{m}^2$ ). **I** The proportion of Synapsin that co-localizes with HOMER1 is 0.51 and the proportion of Synapsin that co-localizes with Gephyrin is 0.21.





**Figure 5. Activity in the Adherent Cortical Organoids**

**A** Snapshot of radially organized neurons transduced with AAV1.Syn.GCaMP6s.WPRE.SV40 (Day 60, 100  $\mu$ m). **B** Representative calcium traces from active neurons in two different Fields of View (FOVs) from cell line 1, showing both individual activity and network bursts where each line corresponds to a single cell. **C** Events per minute of all recorded cells in day 61 and 100 (day 61,  $3.9 \pm 0.5$  events/min; day 100,  $3.9 \pm 0.6$  events/min). **D** Network bursts per minute (D61, average of  $1.4 \pm 0.07$  NB/min; D100 average of  $1.6 \pm 0.12$  NB/min) which shows all active cells are participating with the network bursts to some degree. **E** Percentage of events that are part of network bursts, which indicates that some cells only have events that are part of a network burst while for other cells the network bursts are only a small part of their total calcium events. The data was collected from 2-3 organoids per time point from 2 different batches of differentiation.

screening applications, neurotoxicological studies, mechanistic pathophysiological studies of neurodevelopmental and neuropsychiatric disorders, and pharmacological and phenotypic screening of disease phenotypes during early cortical development.

ACOs also have some limitations, most notably that cortical layering and regional specification appears to be more advanced in free-floating brain organoids, including gyrification in some models. Key characteristics of the model, including concurrent neuronal and astrocyte differentiation and a consistent ACO structure, were robust across hiPSC lines. Because oligodendrocytes arise later and require longer culture times, their generation showed greater variability. We are currently optimizing this aspect of the ACO model. Therefore, the choice of model system should be made based on the specific experimental question being addressed.

Taken together, we present a novel platform for cellular-level human brain modeling using adherent cortical organoids that exhibit high reproducibility and robust neuronal activity. The ability to reliably generate human cortical organoids in multi-well plates combined with neural network functionality offers a unique potential for brain disease modeling and therapeutic screening applications.

## Materials and Methods

### Generation of Neural Progenitor Cells

NPCs from 3 different source hiPSC lines were used. NPC-line 1: in house generated NPCs from human iPSC line WTC11 (Provided by Bruce R. Conklin, The Gladstone Institutes and UCSF, #GM25256, *RRID:CVCL\_Y803* [\[1\]](#), Miyaoka et al., 2014). NPC-line 2: commercially available hNPCs from Axol Biosciences (ax0015). NPC-line 3: NPCs derived using the protocol of Shi et al., (Shi et al., 2012 [\[2\]](#)) from hiPSC line IPSC0028 (Sigma-Aldrich, *RRID:CVCL\_EE38* [\[3\]](#)). Line 1 NPCs were generated from the WTC11 hiPSC line grown on mouse embryonic fibroblasts (MEFs) in human embryonic stem cell medium (Dulbecco's modified Eagle's medium (DMEM)/F12 (Thermo Fisher Scientific), 20% knockout serum (Thermo Fisher Scientific), 1% minimum essential medium/non-essential amino acid (Sigma-Aldrich, St Louis, MO, USA), 71n1ml  $\beta$ -mercaptoethanol (Sigma-Aldrich), 1% L-glutamine (Thermo Fisher Scientific) and 1% penicillin/streptomycin (Thermo Fisher Scientific). The hiPSC colonies were dissociated from the MEFs with collagenase (1001U1ml, Thermo Fisher Scientific, Waltham, MA, USA) for 71min at 371°C/5% CO<sub>2</sub>. Embryoid Bodies (EBs) were generated by transferring dissociated iPSCs to non-adherent plates in human embryonic stem cell medium on a shaker in at 371°C/5% CO<sub>2</sub>. EBs were grown for 2 days in human embryonic stem cell medium, changed into neural induction medium (Advanced DMEM/F12, 1% N2 supplement (Thermo Fisher Scientific), 21 $\mu$ g1ml heparin (Sigma-Aldrich) and 1% penicillin/streptomycin) on day 2 (d2) and cultured for another 5 days in suspension (d3–d7). The EBs were slightly dissociated at d7 by trituration and plated onto laminin-coated 101cm dishes (201 $\mu$ g1ml laminin (Sigma-Aldrich) in DMEM for 301min at 371°C), initially using neural induction medium (d7–d14), and then from d15 in NPC medium consisting of: Advanced DMEM/F12 (Life Technologies), 1% N2 supplement (Thermo Fisher Scientific), 2% B27-RA supplement (Thermo Fisher Scientific), 1  $\mu$ g/ml laminin (L2020, Sigma-Aldrich), 20 ng/ml basic fibroblast growth factor (Merck-Millipore, Darmstadt, Germany) and 1% penicillin/streptomycin (Thermo Fisher Scientific). After passage 3, NPC cultures were purified using fluorescence-activated cell sorting (FACS). NPCs were detached from the culture plate using Accutase (Stem Cell Technologies) and CD184<sup>+</sup>/CD44<sup>+</sup>/CD271<sup>+</sup>/CD24<sup>+</sup> cells (Yuan et al., 2011 [\[4\]](#)) were collected on a FACSaria III Cell Sorter (BD Bioscience) and expanded in NPC medium. NPCs were maintained in NPC medium on 6-well plates (Sanbio, 140675) coated with 20  $\mu$ g/ml laminin (Sigma, L2020) in Advanced DMEM/F12 (Life Technologies) and were used for differentiation to adherent cortical organoids between passage 3 and 7 after the FACS purification step.



## Generating Adherent Cortical Organoids

NPCs were dissociated with Accutase (Stem Cell Technologies), and live cells were counted with Trypan Blue (Stem Cell Technologies) in a Burkert counting chamber. The NPCs were seeded in 384-well plates (M1937-32EA, Life Technologies) that were coated with 50 µg/ml laminin in Neurobasal medium (Thermo Fisher Scientific) for at least 30 minutes at 371°C. The required seeding cell densities to generate ACOs for each individual cell line were determined by seeding a range of cell densities followed by visual inspection with a bright field microscope, Live/Dead staining and immunocytochemistry. When too few NPCs are seeded, the organoids will not form and show patches of clustered cells like you would see in a standard monolayer culture. When too many NPCs are seeded, the 384-plate well will overgrow, leading to a lack of organoid structure formation. The variation in NPC seeding density is mostly explained by NPC proliferation rate, and to a lesser degree by the capacity of a given cell line to differentiate into post-mitotic neural cells. NPCs were seeded and differentiated in Neural Differentiation Medium: Neurobasal medium (Thermo Fisher Scientific), 1% N2 supplement (Thermo Fisher Scientific), 2% B27-RA supplement (Thermo Fisher Scientific), 1% minimum essential medium/non-essential amino acid (Stem Cell Technologies), 201ng/ml brain-derived neurotrophic factor (ProSpec Bio), 201ng/ml glial cell-derived neurotrophic factor (ProSpec Bio), 11µM dibutyl cyclc adenosine monophosphate (Sigma-Aldrich), 2001µM ascorbic acid (Sigma-Aldrich), 21µg/ml laminin (Sigma-Aldrich) and 1% penicillin/streptomycin (Thermo Fisher Scientific). For oligodendrocyte maturation the cells were continuously grown in the presence of 2 ng/ml T3 (Sigma-Aldrich). Every 2-3 days the adherent cortical organoids were refreshed with a 75% media change.

## Immunocytochemistry

For live-dead staining, living cultures were incubated with LIVE/DEAD™ Viability/Cytotoxicity Kit according to manufacturer's instructions (Thermo Fisher Scientific). For immunocytochemistry adherent cortical organoids were fixed for 20-30 minutes using 4% formaldehyde in phosphate-buffered saline (PBS), washed with PBS 3 times and blocked for 1 hour by pre-incubation in staining buffer containing 0.051M Tris, 0.9% NaCl, 0.25% gelatin and 0.5% Triton-X-100 (pH 7.4). Primary antibodies were incubated for 48-72h at 41°C in staining buffer, washed with PBS and incubated with the secondary antibodies in staining buffer for 2h at room temperature. The cultures were embedded in Mowiol 4-88 (Sigma-Aldrich), after which confocal imaging was performed with a Zeiss LSM700 and Zeiss LSM800 confocal microscope using ZEN software (Zeiss, Oberkochen, Germany). The following primary antibodies were used: SOX2 (Merck-Millipore AB5603, 1:200); Nestin (Merck-Millipore MAB5326, 1:200); MAP2 (Synaptic Systems 188004, 1:200); NeuN (Merck ABN78, 1:200); GFAP (Merck-Millipore AB5804, 1:300); FOXG1 (Abcam AB18259, 1:200); CUX1 (Abcam AB54583, 1:200); CTIP2 (Abcam AB18465, 1:100); Synapsin 1/2 (Synaptic Systems 106003, 1:200); and PSD95 (Thermo Fisher Scientific MA1-046, 1:100); HOMER1 (Synaptic Systems 160011, 1:100) Gephyrin (Synaptic Systems 147011, 1:100); (Tau (Cell Signaling Technology 4019, 1:200); S100β (Sigma-Aldrich S2532, 1:200); Pax6 (Santa Cruz sc-81649, 1:100); NG2 (Gift from W. Stallcup Lab, 1:100); NF200 (Sigma-Aldrich 083M4833, 1:200); MBP (Abcam AB7349, 1:100); GFP (Abcam ab13970, 1:100); GAD67 (Merck-Millipore MAB5406, 1:100) DAPI (Thermo Fisher Scientific D1306). The following secondary antibodies were used 1:200: Alexa-488, Alexa-555, Alexa-647 (Jackson ImmunoResearch, West Grove, PA, USA).

## Sparse labelling of excitatory neurons

pENN.AAV9.CamKII.4.eGFP.WPRE.rBG (Addgene viral prep # 105541-AAV9) was added to the cortical organoids at day 278 ( $1.68 \times 10^8$  GC/well). The transduced cortical organoids were fixed and stained on day 310.

## Synaptic Quantification

Following immunocytochemistry confocal imaging was performed with a Zeiss LSM800 confocal microscope with Airy scan using ZEN software (Zeiss, Oberkochen, Germany). Multiple images per organoid were taken of 157.94 µm by 157.94 µm with a depth of 10-15 µm with a scan every 0.17

$\mu\text{m}$ . The surface area of MAP2 and the number of Synapsin+ pre-synaptic terminals quantification was performed on a z-projection of the images in Fiji (Schindelin et al., 2012 [4](#)). The co-localization of Synapsin with either HOMER1 or Gephyrin was manually counted. The Day 114 data in Figure 4H [4](#) and 4I [4](#) was obtained from 4 organoids each with a total area of  $\sim 50000 \mu\text{m}^2$  MAP2 and  $\sim 10000$  Synapsin+ pre-synaptic terminals analysed.

## Calcium imaging

For calcium imaging, the genetically encoded calcium indicator AAV1.Syn.GCaMP6s.WPRE.SV40 (Penn Vector Core, 100843-AAV1) was added to the organoids at day 42 of differentiation ( $1.5 \times 10^8$  GC/well). Recordings were performed at day 60 on a Zeiss LSM800 confocal microscope using ZEN software (Zeiss, Oberkochen, Germany). The recordings were made with a 20x/0.8NA Ph2 Plan-Apochromat objective, with a field of view of  $150 \times 100 \mu\text{m}$  and a pixel size of  $0.3 \mu\text{m}$ . The acquisition rates of the recordings were between 4-5 f.p.s. 24 hours before the recordings, the medium was switched to BrainPhys Neural Differentiation Medium. BrainPhys Neuronal Media (Stem Cell Technologies), 1% N2 supplement (Thermo Fisher Scientific), 2% B27-RA supplement (Thermo Fisher Scientific), 1% minimum essential medium/non-essential amino acid (Stem Cell Technologies), 1% penicillin/streptomycin (Thermo Fisher Scientific), 201ng/ml brain-derived neurotrophic factor (ProSpec Bio), 201ng/ml glial cell-derived neurotrophic factor (ProSpec Bio),  $1 \mu\text{M}$  dibutyl cyclic adenosine monophosphate (Sigma-Aldrich),  $2001 \mu\text{M}$  ascorbic acid (Sigma-Aldrich) and  $2 \mu\text{g/ml}$  laminin (Sigma-Aldrich). The calcium imaging recordings were processed using CNMF-E (Pnevmatikakis et al., 2016) and FluoroSNAPP (Patel et al., 2015 [4](#)) to generate the traces. Calcium traces were then analysed using a custom script for event and network burst detection using an algorithm written in Python (v3.8.2) (code accessibility can be requested via Github). A network burst is defined here as synchronous calcium events of at least 60% of the active neurons in the field of view.

## Cyquant Proliferation Assay

CyQUANT™ Direct Cell Proliferation Assay, C35011 (Thermo Fisher Scientific) was used according to manufacturer specifications. For each time point and each of the 3 NPC lines 10-12 wells of a 96-well plate were seeded with NPCs (2500 NPCs per well). In addition, NPC lines generated from 2 different clones from IPS line MH0159020 (Rutgers University Cell and DNA Repository) were added to increase the dynamic range of the proliferation curve. At 24h and 96h NPCs were frozen at  $-80^\circ\text{C}$ . All NPC lines and timepoints were thawed, lysed and measured together. Doubling time was calculated between 24h and 96h. None of the wells were confluent at 96h.

## Statistical analysis

All data represent mean  $\pm$  SEM. When comparing developmental markers in Figure S2 we used one-way ANOVA followed by Tukey-Kramer's multiple correction test.  $n=3-6$  images taken over two wells, for each time point. To compare the time points in Figure 5C-E [4](#) we used a Mann-Whitney test.

## Data availability

All data are available on request

## Acknowledgements

This work was supported by the Netherlands Organ-on-Chip Initiative, an NWO Gravitation project funded by the Ministry of Education, Culture and Science of the government of the Netherlands (024.003.001) to SAK, Hersenstichting Fellowship (F2012(1)-39) to FMSdV, an Erasmus MC Human Disease Model Award to FMSdV and a Dutch ZonMw MKMD Create2Solve grant (114025201) to SAK and FMSdV.



## Additional information

### Author contributions

MvdK, SB and HS performed experiments, MU developed calcium imaging analysis algorithms, FdV and SAK supervised the work, MvdK, FdV and SAK wrote the paper.

### Funding

| Funder                                                               | Grant reference number | Author                                                  |
|----------------------------------------------------------------------|------------------------|---------------------------------------------------------|
| Nederlandse Organisatie voor Wetenschappelijk Onderzoek (NWO)        | 024.003.001            | Steven A Kushner<br>Femke MS de Vrij<br>Maurits A Unkel |
| Hersenstichting (Hersenstichting Nederland)                          | (F2012(1)-39           | Femke MS de Vrij                                        |
| ZonMw (Netherlands Organisation for Health Research and Development) | 114025201              | Steven A Kushner<br>Femke MS de Vrij                    |

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## Additional files

**Supplementary figure 1.** [↗](#) Reproducibility of Neural Progenitor Cells and Adherent Cortical Organoids. **A** NPCs are positive for SOX2, Nestin, FOXG1, PAX6 and TBR2 (scale bars 20  $\mu$ m) **B** LIVE/DEAD staining of adherent organoids (N=4) showing reproducibility of the structure formation. Seeding density line 1; 1250 NPCs, line 2; 750 NPCs, line 3; 1000 NPCs (scale bars 500  $\mu$ m). **C** Proportion of properly formed single structure adherent cortical organoids. Each dot represents the end point of a different batch of adherent cortical organoids where the size of the batch ranges from 10-40 organoids per plate. Total organoids line 1 n=248, line 2 n=70, line 3 n=70. **D** Function of the necessary NPC seeding density to form adherent cortical organoids in relation to the doubling time as a measure of the proliferation rate of the NPCs. The doubling time of the NPCs explains more than half the variation ( $r = 0.67$ ) of the required seeding density showing that fewer cells need to be seeded for NPC lines with a higher proliferation rate.

**Supplementary figure 2.** [↗](#) Neuronal maturation in cortical organoids is shown by a temporal shift in NPC and neuronal markers. **A** Representative images exhibiting change in expression of NPC and neuronal markers over time. Number of cells positive for NPC markers SOX2 and PAX6 decreases over time. Deep-layer CTIP2+ neurons start appearing at day 28, while upper-layer CUX1+ positive neurons appear in higher numbers around day 56. (Scale bars 20  $\mu$ m). **B** The percentages of DAPI+ cells expressing each of the markers were quantified at four time points. Significant differences between week 2 and week 8 expression were seen for SOX2+ (down by 39.6 %), CTIP2+ (up by 13.5 %) and CUX1+ cells (up by 23.1 %). \*\*\*  $p < 0.0001$ , ANOVA one-way followed by Tukey- Kramer's multiple correction test. Error bars indicate SEM, n=3-6 images taken over two wells, for each time point.

**Supplementary figure 3.** [↗](#) Distribution of axons and dendrites in adherent cortical organoids. Radially organized MAP2+/Tau+ dendrites along with both radially and circumferentially organized NF200+/Tau+ axons in duplicate for each cell line (Day 63, scale bars 500  $\mu$ m)

**Supplementary figure 4.** [↗](#) Cortical layering in the adherent cortical organoids. **A,B** Separation of CUX1 and CUX2 expression (A day 70, scale bar 100  $\mu$ m; B day 67, scale bar 500  $\mu$ m) **C** A SOX2+ band/layer is visible in the centre of the ACO (day 63, scale bar 500  $\mu$ m)

**Supplementary figure 5.** [↗](#) Astrocyte distribution within Adherent Cortical Organoids. The abundantly present GFAP+ and S100B+ astrocytes have a comparable distribution with the MAP2+ neurons in the ACOs. The astrocytes show radial patterns with many somas located in the centre

as well as a large presence outside the centre (Day 63, scale bars 500  $\mu\text{m}$ ).

**Figure 5—Video 1.** [Video showing 10 minutes of calcium imaging of an ACO at Day 61 using the genetically encoded calcium indicator AAV1.Syn.GCaMP6s.WPRE.SV40 \(rainbow filter indicates increasing calcium signal ranging from blue to red, speed increase indicated by time stamp in top left corner\).](#)

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## Peer reviews

### Reviewer #3 (Public review):

#### Summary:

Kroeg et al. introduced a novel method for generating 3D cortical layer-like organization in hiPSC-derived models, achieving remarkably consistent topography within compact dimensions. Their approach involves seeding frontal cortex-patterned iPSC-derived neural progenitor cells into 384-well plates, which triggers the spontaneous assembly of adherent cortical organoids comprising diverse neuronal subtypes, astrocytes, and oligodendrocyte-lineage cells.

#### Strengths:

Compared with existing brain organoid models, these adherent cortical organoids exhibit enhanced reproducibility and improved cell viability during prolonged culture, thereby providing versatile opportunities for high-throughput drug discovery, neurotoxicological screening, and investigation of brain disorder pathophysiology. Overall, this study addresses an important and timely need for advancing current brain organoid systems.

#### Weaknesses:

Highlighting the consistency of differentiation across different cell lines and standardizing functional outputs are crucial to emphasize the broad future potential of this new organoid system for large-scale pharmacological screening. The authors provided a substantial amount of new data during the revision process to support the reproducibility of neuronal activity.

The next step would be to leverage this platform for functional screening of chemical and genetic perturbations to identify new drug candidates.

Comments on revisions:

Most of my previous concerns were adequately addressed through the revision.

<https://doi.org/10.7554/eLife.98340.2.sa1>

## Author response:

The following is the authors' response to the original reviews.

### Public Reviews:

#### Reviewer #1 (Public review):

##### Summary:

*Kroeg et al. describe a novel method for 2D culture human induced pluripotent stem cells (hiPSCs) to form cortical tissue in a multiwell format. The method claims to offer a significant advancement over existing developmental models. Their approach allows them to generate cultures with precise, reproducible dimensions and structure with a single rosette; consistent geometry; incorporating multiple neuronal and glial cell types (cellular diversity); avoiding the necrotic core (often seen in free-floating models due to limited nutrient and oxygen diffusion). The researchers demonstrate the method's capacity for long-term culture, exceeding ten months, and show the formation of mature dendritic spines and considerable neuronal activity. The method aims to tackle multiple key problems of in vitro neural cultures: reproducibility, diversity, topological consistency, and electrophysiological activity. The authors suggest their potential in high-throughput screening and neurotoxicological studies.*

##### Strengths:

*The main advances in the paper seem to be: The culture developed by the authors appears to have optimal conditions for neural differentiation, lineage diversification, and long-term culture beyond 300 days. These seem to me as a major strength of the paper and an important contribution to the field. The authors present solid evidence about the high cell type diversity present in their cultures. It is a major point and therefore it could be better compared to the state of the art. I commend the authors for using three different IPS lines, this is a very important part of their proof. The staining and imaging quality of the manuscript is of excellent quality.*

We thank the reviewer for the positive comments on the potential of our novel platform to address key problems of *in vitro* neural culture, highlighting the longevity and reproducibility of the method across multiple cell lines.

##### Weaknesses:

*(1) The title is misleading: The presented cultures appear not to be organoids, but 2D neural cultures, with an insufficiently described intermediate EB stage. For nomenclature, see: doi: 10.1038/s41586-022-05219-6. Should the tissue develop considerable 3D depth, it would suffer from the same limited nutrient supply as 3D models - as the authors point out in their introduction.*

We appreciate the opportunity to clarify this point. We respectfully disagree that the cultures do not meet the consensus definition of an organoid. In fact, a direct quote from the seminal nomenclature paper referenced by the reviewer states: "We define organoids as *in vitro*-



generated cellular systems that emerge by self-organization, include multiple cell types, and exhibit some cytoarchitectural and functional features reminiscent of an organ or organ region. Organoids can be generated as 3D cultures or by a combination of 3D and 2D approaches (also known as 2.5D) that can develop and mature over long periods of time (months to years).” (Pasca et al, 2022 doi10.1038/s41586-022-05219-6). Therefore, while many organoid types indeed have a more spherical or globular 3D shape, the term organoid also applies to semi-3D or nonglobular adherent organoids, such as renal (Czerniecki et al 2018, doi.org/10.1016/j.stem.2018.04.022) and gastrointestinal organoids (Kakni et al 2022, doi.org/10.1016/j.tibtech.2022.01.006). Accordingly, the adherent cortical organoids described in the manuscript exhibit self-organization to single radial structures consisting of multiple cell layers in the z-axis, reaching ~200um thickness (therefore remaining within the limits for sufficient nutrient supply), with consistent cytoarchitectural topology and electrophysiological activity, and therefore meet the consensus definition of an organoid.

*(2) The method therefore should be compared to state-of-the-art (well-based or not) 2D cultures, which seems to be somewhat overlooked in the paper, therefore making it hard to assess what the advance is that is presented by this work.*

It was not our intention to benchmark this model quantitatively against other culture systems. Rather, we have attempted to characterize the opportunities and limitations of this approach, with a qualitative contrast to other culture methods. Compared to state-of-the-art 2D neural network cultures, adherent cortical organoids provide distinct advantages in:

- (1) Higher order self-organized structure formation, including segregation of deeper and upper cortical layers.
- (2) Longevity: adherent cortical organoids can be successfully kept in culture for at least 1 year, whereas 2D cultures typically deteriorate after 8-12 weeks.
- (3) Maturity, including the formation of dendritic mushroom spines and robust electrophysiological activity.
- (4) Cell type diversity including a more physiological ratio of inhibitory and excitatory neurons (10% GAD67+/NeuN+ neurons in adherent cortical organoids, vs 1% in 2D neural networks), and the emergence of oligodendrocyte lineage cells.

On the other hand, limitations of adherent cortical organoids compared to 2D neural network cultures include:

- (1) Culture times for organoids are much longer than for 2D cultures and the method can therefore be more laborious and more expensive.
- (2) Whole cell patch clamping is not easily feasible in adherent cortical organoids because of the restrictive geometry of 384-well plates.

*(3) Reproducibility is prominently claimed throughout the manuscript. However, it is challenging to assess this claim based on the data presented, which mostly contain single frames of unquantified, high-resolution images. There are almost no systematic quantifications presented. The ones present (Figure S1D, Figure 4) show very large variability. However, the authors show sets of images across wells (Figure S1B, Figure S3) which hint that in some important aspects, the culture seems reproducible and robust.*

We made considerable efforts to establish quantitative metrics to assess reproducibility. We applied a quantitative scoring system of single radial structures at different time points for multiple batches of all three lines as indicated in Figure S1C. This figure represents a comprehensive dataset in which each dot represents the average of a different batch of organoids containing 10-40 organoids per batch. To emphasize this, we have adapted the

graph to better reflect the breadth of the dataset. Additional quantifications are given in Figure S2 for progenitor and layer markers for Line 1 and in Figure 2 for interneurons across all three lines, showing relatively low variability. That being said, we acknowledge the reviewer's concerns and have modified the text to reduce the emphasis of this point, pending more extensive data addressing reproducibility across an even broader range of parameters.

*(4) What is in the middle? All images show markers in cells present around the center. The center however seems to be a dense lump of cells based on DAPI staining. What is the identity of these cells? Do these cells persist throughout the protocol? Do they divide? Until when? Addressing this prominent cell population is currently lacking.*

A more comprehensive characterization of the cells in the center remains a significant challenge due to the high cell density hindering antibody penetration. However, dyebased staining methods such as DAPI and the LIVE/DEAD panel confirm a predominance of intact nuclei with very minimal cell death. The limited available data suggest that a substantial proportion of the cells in the center are proliferative neural progenitors, indicated by immunolabeling for SOX2 (Figure 2A,D;Figure S4C). Furthermore, we are currently optimizing the conditions to perform single cell / nuclear RNA sequencing to further characterize the cellular composition of the organoids.

*(5) This manuscript proposes a new method of 2D neural culture. However, the description and representation of the method are currently insufficient. (a) The results section would benefit from a clear and concise, but step-by-step overview of the protocol. The current description refers to an earlier paper and appears to skip over some key steps. This section would benefit from being completely rewritten. This is not a replacement for a clear methods section, but a section that allows readers to clearly interpret results presented later.*

We have revised the manuscript to include a more detailed step-by-step overview of the protocol.

*(b) Along the same lines, the graphical abstract should be much more detailed. It should contain the time frames and the media used at the different stages of the protocol, seeding numbers, etc.*

As suggested, we have adapted the graphical abstract to include more detail.

#### **Reviewer #2 (Public review):**

##### *Summary:*

*In this manuscript, van der Kroeg et al have developed a method for creating 3D cortical organoids using iPSC-derived neural progenitor cells in 384-well plates, thus scaling down the neural organoids to adherent culture and a smaller format that is amenable to high throughput cultivation. These adherent cortical organoids, measuring 3 x 3 x 0.2 mm, self-organize over eight weeks and include multiple neuronal subtypes, astrocytes, and oligodendrocyte lineage cells.*

##### *Strengths:*

*(1) The organoids can be cultured for up to 10 months, exhibiting mature dendritic spines, axonal myelination, and robust neuronal activity.*

*(2) Unlike free-floating organoids, these do not develop necrotic cores, making them ideal for high-throughput drug discovery, neurotoxicological screening, and brain disorder studies.*

*(3) The method addresses the technical challenge of achieving higher-order neural complexity with reduced heterogeneity and the issue of necrosis in larger organoids. The method presents a technical advance in organoid culture.*

*(4) The method has been demonstrated with multiple cell lines which is a strength.*

*(5) The manuscript provides high-quality immunostaining for multiple markers.*

We appreciate the reviewer's acknowledgement of the strengths of this novel platform as a technical advance in organoid culture that reduces heterogeneity and shows potential for higher throughput experiments.

*Weaknesses:*

*(1) Direct head-to-head comparison with standard organoid culture seems to be missing and may be valuable for benchmarking, ie what can be done with the new method that cannot be done with standard culture and vice versa, ie what are the aspects in which new method could be inferior to the standard.*

In our opinion, it would be extremely difficult to directly compare methods. Most notably, whole brain organoids grow to large and irregular globular shapes, while adherent cortical organoids have a more standardized shape confined by the geometry of a 384well. Moreover, it was not our intention to benchmark this model quantitatively against other culture systems. Rather, we have attempted to characterize the opportunities and limitations of this approach, with a qualitative contrast to other culture methods, as addressed in response to comment 2 of Reviewer 1 above.

*(2) It would be important to further benchmark the throughput, ie what is the success rate in filling and successfully growing the organoids in the entire 384 well plate?*

Figure S1 shows the success rate of organoid formation and stability of the organoid structures over time. In addition, we have added the number of wells that were filled per plate.

*(3) For each NPC line an optimal seeding density was estimated based on the proliferation rate of that NPC line and via visual observation after 6 weeks of culture. It would be important to delineate this protocol in more robust terms, in order to enable reproducibility with different cell lines and amongst the labs.*

Figure S1 provides the relationship between proliferation rate and seeding density, allowing estimation of seeding densities based on the proliferation rate of the NPCs. However, we appreciate the reviewers' feedback and have modified the methods to provide more detail.

### **Reviewer #3 (Public review):**

*Summary:*

*Kroeg et al. have introduced a novel method to produce 3D cortical layer formation in hiPSC-derived models, revealing a remarkably consistent topography within compact dimensions. This technique involves seeding frontal cortex-patterned iPSC-derived neural progenitor cells in 384-well plates, triggering the spontaneous assembly of adherent cortical organoids consisting of various neuronal subtypes, astrocytes, and oligodendrocyte lineage cells.*

*Strengths:*

*Compared to existing brain organoid models, these adherent cortical organoids demonstrate enhanced reproducibility and cell viability during prolonged culture,*



*thereby providing versatile opportunities for high-throughput drug discovery, neurotoxicological screening, and the investigation of brain disorder pathophysiology. This is an important and timely issue that needs to be addressed to improve the current brain organoid systems.*

We thank the reviewer for highlighting the strengths of our novel platform. We appreciate that all three reviewers agree that the adherent cortical organoids presented in this manuscript reliably demonstrate increased reproducibility and longevity. They also commend its potential for higher throughput drug discovery and neurotoxicological/phenotype screening purposes.

*Weaknesses:*

*While the authors have provided significant data supporting this claim, several aspects necessitate further characterization and clarification. Mainly, highlighting the consistency of differentiation across different cell lines and standardizing functional outputs are crucial elements to emphasize the future broad potential of this new organoid system for large-scale pharmacological screening.*

We appreciate the feedback and have added more detail on consistency and standardization of functional outputs.

**Recommendations for the authors:**

**Reviewer #1 (Recommendations for the authors):**

*Minor points*

*(1) As the preprint is officially part of the eLife review, I have to remark that the preprint which is made available on bioRxiv, suffers from some serious compatibility or format problem: one cannot highlight sentences as in a regular PDF and when trying to copy paste sentences from it jumbled characters are copied to the clipboard.*

The updated version of the paper on *bioRxiv* should not suffer from these compatibility issues.

*(2) Since the paper is presenting a new method it should briefly describe how each step, including the hiPSC culture was done, the reference to an earlier publication in this case is not sufficient, and this practice is generally best to avoid for methods papers.*

Each step in the culturing process has now been described in the methods.

*(3) The EB stage is insufficiently described. The "2D - 3D - 2D" transitions should be clearly explained.*

The methods section has been rewritten and expanded to include these processes in more detail.

*(4) Is there one FACS sorting in the protocol, or multiple (additional at IPS culture)? What markers each? What is the motivation for sorting and purifying the neural progenitors? Was the culture impure? What was purity? What cell types are expected after sorting, and what is removed?*

Only one FACS sorting step is performed at the NPC stage. This was added as an improvement to our original neural network protocol (Günhanlar et al 2018) to ensure consistency over different hiPSC source cell lines that can yield variable amounts of frontal cortical patterned NPCs. Positive sorting for neural lineage markers CD184 and CD24, and negative sorting for mesenchymal/neural crest CD217 and CD44 glial progenitor markers, according to Yuan et al 2011, ensures frontal-patterned cortical NPCs as confirmed for all batches by

immunohistochemistry for SOX2, Nestin and FOXG1. We have added new text to the Methods section to clarify this more explicitly.

*(5) Seeding protocol and parameters are insufficiently described, and from what I read they are poorly defined: "Specifically, the optimal seeding density was determined by visual inspection of the organoids between 28 to 42 days after seeding a range of cell densities in the 384-well plate wells." For a new method, precise, actionable instructions are needed. I may have overlooked those elsewhere, in this case, please clarify these sections.*

The Methods section was rewritten and expanded to describe the methodology in greater detail with more actionable instructions.

*(6) The timeline in Figure 1 is not clearly delineated; I found it hard to understand which figure corresponds to which stage (e.g. facs sorting is not mentioned in the first part of the results but it is part of Figure 1A, neural rosette formation can happen both before and after facs sorting, simply referring to rosettes is not clear). Later parts of the manuscript clearly introduce the terms sorting and seeding in the context of this method, and how ages (days) refer to these time points.*

Figure 1 was adapted to clarify the generation of Neural Progenitor Cells (NPCs) and subsequent seeding of NPCs to generate Adherent Cortical Organoids (ACOs).

*(7) The authors define: "cortical organized defined as a single radial structure." This is not a commonly used definition of organoids, for nomenclature, please see: doi: 10.1038/s41586-022-05219-6 (Pasca et al 2022).*

To clarify, the statement is not meant to reflect a definition of organoids in general, but rather the scoring of proper structure formation for Figure S1C. For discussion on nomenclature, see our response to point 1 of Reviewer 1 in the public review. We changed the wording to be more accurate.

*(8) In Figure S1d, the authors write: "the fraction of structurally intact cultures decreased to 50%", but I'm looking at that graph there seems to be no notable decrease, but huge variability. The authors should quantify claims of decrease by linear regression and an R square. Variation within and the cross-cell lines seem to be large. Also, it is unclear if dots are corresponding to the same wells/plates, in other words: is this a longitudinal experiment? What is the overall success rate? How is success determined? Are there clear criteria? to the same wells/plates, in other words: is this a longitudinal experiment? What is the overall success rate? How is success determined? Are there clear criteria?*

We agree with the reviewer that the claim on fraction of intact cultures decreasing over time to 50% is an overinterpretation due the large variability. We changed the wording in the manuscript to: While some later batches show moderately reduced success rates compared with the earliest batches, properly formed single-structure organoids were still obtained at 40–90% success across all examined time points (Figure S1C), indicating that long-term culture is feasible albeit with variable efficiency. The data are not longitudinal as each dot represents an endpoint of a different batch of organoids, totaling 18 independent batches across the three lines. We have clarified this in the figure legend. Success was defined at the well level as the presence of a single, continuous radial structure occupying the well, without obvious fragmentation or fusion events, as assessed by LIVE/DEAD that also confirmed viability. Wells were scored as successful only when the radial structure showed predominantly live signal with no large necrotic areas. Wells containing multiple radial structures, fused aggregates, or predominantly dead tissue were scored as unsuccessful.

*(9) Figure s1c: the numbering to this panel should be swapped, because it is referenced after other panels in the text. The reference is confusing: "Plotting the interaction*

*between proliferation and the amount of NPCs required to be seeded for the successful generation of adherent cortical organoids" - success is not present in this graph at all? How is that measured?*

Figures S1C and S1D have been adapted to clarify the measure of ‘successful organoid formation’.

*(a) The description of this plot is confusing: "The doubling time of the NPCs explains more than half the variation ( $r^2 = 0.67$ ) of the required seeding density." What else is there? I thought that this was the formula the authors suggested to determine seeding density, but it seems not. Or is "manual inspection" the determinant, and that seems to correlate with this metric?*

Even though the rate of proliferation, measured as doubling time, is the main determinant of the seeding density, it is not the only determinant of the seeding density. For instance, intrinsic differences in differentiation potential could also play a role. Therefore, NPC lines with similar doubling times might still have slightly different optimal seeding densities. We have added clarification of this conclusion to the Results section.

*(b) Seeding density is a key parameter in many in vitro differentiation and culture protocols. This importance however does not mean that this density is attributable to differences in cell proliferation rate. Alternatively, the amount of cells determines the amount of secreted molecules and cell-to-cell contacts.*

Here, when we refer to the cell density, we specifically refer to the cell density needed to generate the ACO. We show that the most important contributor to the variation in ACO formation is the proliferation, measured here as the doubling time. We agree that there are other factors involved such as the secreted molecules, cell-to-cell contacts as well as the ability of a given NPC line to differentiate into a post-mitotic cell.

*(c) Is it mentioned which cell line this experiment corresponds to?*

The data in Figure S1D is from the 3 reported cell lines, as well as 2 clones from a fourth IPS cell line. This is detailed in the Methods section of the proliferation assay.

*(d) Without a more detailed explanation, seeding density and doubling time could be independent variables.*

These two variables are highly correlated as shown in Figure S1D, but it is true that there can be other variables that account for the observed variance, as discussed above in Point 9b.

*(e) In this figure the success rate is not visible at all so I have no idea how the authors arrive at a conclusion about success rate.*

We have adapted the figure legend to reflect which cell lines the dots in Fig. S1D represent. NPC lines can have substantial variation in proliferation rates. The figure reflects data of NPCs of 5 clones of 4 different hiPSC lines (as indicated in the Methods) with different proliferation rates. Also, the ACO success rate (operationally defined uniformly to the data shown in Fig. S1C) was also included.

*(10) Figure 2: Clean spatial segregation seems to be a strength of the system and therefore I would recommend putting more of the relevant microscopy images to the main figure, which are now currently in Figure S4.*

We have adapted Figure 2 accordingly, and included additional representative cortical layering images in Figure S4.

*(11) The variability in interneuron content seems to be significant, as currently presented in the figure. However, this may be due to a special organization. It would first quantify in consecutive rings around the centers whether interneurons have a tendency to be enriched towards the center or the edge of the culture. Maybe this explains the variability that is currently present in Figure s5b.*

We agree that spatial organization of interneurons could, in principle, contribute to variability. In our analysis, however, images were acquired from positions selected by a random sampling grid across the entire culture, rather than from specific central or peripheral regions. Each field contained on average  $130.6 \pm 16.1$  NeuN+ nuclei, which provided a relatively large sampling volume per position. If interneurons were strongly enriched at the center or edge, we would expect systematic differences in interneuron fraction between fields assigned to central versus peripheral grid positions. We did not observe such a pattern in our dataset, suggesting that spatial organization is not the main driver of the observed variability.

*(12) Because in previous figures it seems like there is considerable variability across individual cultures and images here are coming from separate cultures, please use different shapes of the points coming from different cultures/wells, to see if maybe there is a culture-to-culture difference that explains the variability present in the figure.*

We have added different symbols per organoid for the interneuron quantifications and moved this quantification to main Figure 2.

*(13) I believe it is currently the standard error of the mean which is displayed in the figure, which is not an appropriate representation for variability, or the reproducibility across individual data points. SEM quantifies the reproducibility of the mean, not the reproducibility of the individual data points, which matters here. Mean refers to the mean of this quantification experiment and therefore it's not a biological entity. A box plot showing the interquartile range besides the individual data points would be an accurate representation of the spread of the data.*

We agree and have adapted the data, now in Figure 5, accordingly.

*(14) Again, in general, the main figures should contain much more of the quantification, as opposed to just raw images.*

Quantifications have been added in Figure 2 for the GAD67/NeuN for all cell lines as well as a time course quantification of GAD67/NeuN for 1 of the cell lines. In Figure 4, we have added excitatory and inhibitory synaptic quantifications.

*(15) Figure 2F-I the location of the center of the rosette should be marked with a star so that the conclusion about the direction of processes can be established.*

The suggested addition of a marker at the center of each rosette was evaluated but not implemented, because it reduced rather than improved figure clarity.

*(16) Figure 3 b and c:*

*High magnification images of single cells, can't show changes in cell type morphology, and one cannot conclude that these cells are present in significant numbers across time. Zoomed-out images or quantification would be necessary for such a claim. The authors already have such images as presented in the next panels, so quantification without new experiments. I am uncertain about the T3 supplement here - do these images correspond to the same conditions?*



*(a) It is unclear to me why different markers are used in the different panels, namely why NG2 is not used in any of the other images.*

NG2 was used at early developmental time points to show the presence of Oligodendrocyte Precursor Cells (OPCs). At later time points, the focus switched to MBP staining to indicate more mature oligodendrocyte lineage cells. Although NG2 and MBP are not in the same panels, the staining was performed for both antibodies at the same developmental time point (Day 119) as seen in Figure 3C and 3D.

*(b) Color coding in Figure 3G is ambiguous; the use of two blues should be avoided, and the Sub-sub panels should be individually labeled for the color code.*

We agree, and have now used different colors.

*(c) It is unclear if the presence of the t3 molecule is part of the standard procedure or if it was a side experiment to enhance the survival of oligodendrocytes. Are there no oligodendrocytes without? How does T3 affect other cell types, and the general health and differentiation of the cultures?*

Indeed, T3 is essential for oligodendrocyte formation. We did not observe obvious effects on the general health or differentiation potential of the cultures.

*(d) Is the 2ng/ml t3 from day one to the final day?*

Indeed, in the organoids cultured to study oligodendrocyte formation, T3 was added from Day 1. These details have now been clarified in the Methods and Results sections.

*(17) Figure 4:*

*(a) Microscopy in this figure is high quality and very convincing about neural maturity.*

*(b) The term "cluster" should be avoided. Unclear what it means here, but my best guess is "cells in a frame of view." Cluster is used with a different meaning in electrophysiology.*

This was adapted to 'neurons in a field of view (FOV)'.

*(c) Panel J: I assume each row corresponds to a single cell? Could this be clarified? Are these selected cells from each frame, or all active cells are represented?*

Indeed, each row corresponds to a single cell, showing all active cells in the frame. This is now clarified in the legend.

*(d) How many Wells do these data correspond to, and in which line it was measured?*

As reported in the legend for Figure 5, these data correspond to 2 wells at Day 61 to which we have now added calcium imaging data from 3 wells from a different batch at Day 100. We have included in the legend that these recordings were from Line 1.

*(e) Panels G to I, again, the use of standard error of the mean is inappropriate and misleading: looking at the error bar one must conclude that there is minimal variation, which is the exact opposite of the conclusions, when one would look at the variability of the raw data points.*

As suggested, the graphs have been adapted as boxplots with interquartile ranges to highlight the distribution of data points.

*(f) It is unclear how many neurons and how many total actively firing neurons are present in the videos analyzed*

All neurons that were active in the field of view and showed at least one calcium event during the ~10 minute recording were included in the analysis. Using this method, we cannot comment on the proportion of neurons that were active from the total amount of neurons present, since the AAV virus we used does not transduce all neurons.

*(g) This figure shows the strength of the method in achieving neural maturity and function. There seems to be that there is considerable activity in the neuronal cultures analyzed. To conclude how reliably the method leads to such mature cultures one would need to measure at least a dozen wells (even if with some simpler and low-resolution method). Concluding reproducibility from one or two hand-picked examples is not possible.*

We agree with the reviewer that the number of wells used for calcium imaging analysis was limited. We are currently working on more advanced methods to increase the throughput of this analysis. However, we've now added another timepoint to the calcium imaging data in Figure 5 from an independent batch of 3 adherent cortical organoids, which demonstrates continued robust activity at Day 100, as well as Day 61.

*Methods:*

*(1) Stem cell culture. The artist described that line 3 is grown on MEFs. Is this true for the other two lines, furthermore were they cultured in identical conditions?*

Line 2 and 3 were not grown on MEFs. We specifically chose different sources of NPCs to reflect the robust nature of the differentiation protocol. We have recently also adapted the protocol from Line 3 NPCs to confirm that the protocol also works starting from hiPSCs grown in feeder-free conditions in StemFlex medium, by adapting NPC differentiation according to our recent publication in *Frontiers in Cellular Neuroscience* (Eigenhuis et al 2023).

*(2) "NPCs were differentiated to adherent cortical organoids between passages 3 and 7 after sorting." Please clarify this sentence. I assume it refers to the first facs sorting of the protocol, but a section is not sufficiently detailed.*

We have adapted the methods to clarify that the FACS purification step occurs at the NPC stage.

*(3) I didn't fully understand: It seems to be that there are two steps of fact sorting involved, one after passage 3 and one after week 4. This should be represented in the graphical abstract of Figure 1.*

As outlined above, there is only 1 FACS sorting step at NPC stage. We have adapted this in the Methods and in the graphical abstract.

*(4) Neural differentiation: The authors write that optimal seeding density was determined by visual inspection of the organoids - this is.*

We have clarified the Methods section to better explain the process of optimizing the seeding density for each NPC line to generate the ACOs.

*(5) What does the following sentence mean: "Cells were refreshed every 2-3 days." Does it mean in replacement of the complete media? How much Media was added to the Wells?*

This is a very good point that we have now clarified in the Methods, as full replenishment of media is neither feasible, nor desirable. From the total volume of 110  $\mu$ l per well, 80  $\mu$ l is taken out and replaced with 85  $\mu$ l to compensate for evaporation.

*(6) Calcium imaging: can the authors explain the decision to move the cultures one day before imaging into brainphys neural differentiation medium? In 3D organoid protocols,*

*brainphys is gradually introduced to avoid culture shock (very different composition), and used for multiple months to enhance neural differentiation. For recording electrophysiological activity, artificial CSF is the most common choice.*

Indeed, for whole cell recordings of 2D neural networks as performed in Günhanlar et al 2018, we used gradual transition to aCSF. For the current ACOs, we found that using BrainPhys from the start of organoid differentiation prevents structure formation, probably because of increased speed of maturation disrupting proliferation and organization of radial glia differentiation. However, by changing the media to BrainPhys just one day before recording (reflecting a gradual change as not all medium is fully replenished and easier than switching to aCSF during recording), we saw greatly improved neuronal activity.

*(7) Statistical analysis : As I pointed out before, the standard error of the mean is not an appropriate metric to represent the variability of the data. It is meant to represent the variability of the estimated average. The following thought experiment should make it clear: I measured the expression of a gene in my system. 50 times I measured 0 and 50 times I measured 100. The average is 50, but of course it is a very bad representation of the data because no such data points exist with that value. Yet the standard error of the mean would be plus minus 5.*

We have revised Figures 5C–5D to boxplots displaying the interquartile range with all individual data points overlaid, which more accurately represents the variability in the dataset.

#### Discussion

*(1) The discussion focuses on human cortical development, however, the methods presented by the authors entail dissociation and replating through multiple stages not part of brain development. I see the approach as more valuable as a possibly reliable method that generates both diverse and mature neural cultures.*

We have revised the Discussion to avoid explicitly invoking an *in vitro* recapitulation of human cortical development. Nevertheless, given that the NPCs from which the organoids originate exhibit frontal cortical identity, coupled with the timely emergence of cortical neuronal markers and rudimentary cortical layering, we are increasingly confident that the development of these cultures most likely mirrors that of the frontal cortex. To further substantiate this hypothesis, single-cell RNA sequencing experiments will be conducted in the future to provide additional insights.

*(2) One of the major claims of the authors is that the method is very reproducible. However, there is almost no data on reproducibility throughout the paper. Mostly single, high magnification images are presented, which therefore represent a small region of a single well of a single batch of a single cell line. Based on the data presented it is not possible to evaluate the reproducibility of the method.*

We agree that the original version did not sufficiently document reproducibility. To address this, we have refined and expanded our presentation of reproducibility data. The previous success-rate panel (original Figure S1D) has been moved and adapted as the new Figure S1C. In this updated version, each dot still represents the endpoint success rate of an independent batch, but dot size now scales with batch size (10–40 organoids), and the legend specifies the total numbers of organoids analyzed per line (line 1: n=248; line 2: n=70; line 3: n=70). Together with the distribution of success rates between ~40–90% across multiple time points and three iPSC lines, this more detailed representation allows readers to directly assess the robustness of line-to-line and batch-to-batch performance. In addition, new time course quantifications of interneuron proportion (Figure 2G,H), synaptic marker densities (Figure 4H, I), and late-stage calcium imaging (Figure 5C,D,E) further demonstrate that key structural

and functional read-outs show overlapping ranges across lines and independent differentiations, reinforcing that the method yields reproducible core phenotypes despite some biological variability.

*(3) The data presented is very promising, and it suggests that the authors derived optimal conditions for neural differentiation and neural culture diversification. I am confident that the authors can show that reproducibility, at least in a practical sense (e.g. in wells that form a culture) is high.*

*Overall, this is a very promising and exciting work, that I am looking forward to reading in a mature manuscript.*

**Reviewer #2 (Recommendations for the authors):**

*(1) Direct head-to-head comparison with standard organoid culture seems to be missing and may be valuable for benchmarking, ie what can be done with the new method that cannot be done with standard culture and vice versa, ie what are the aspects in which new method could be inferior to the standard.*

We have now more clearly elaborated the differences with other methods. As addressed in our response to point 2 of Reviewer 1 in the public reviews, there are several limitations and advantages to the adherent cortical organoids model listed as follows:

Advantages of adherent cortical organoids:

- (1) Higher order self-organized structure formation, including segregation of deeper and upper cortical layers.
- (2) Longevity: adherent cortical organoids can be successfully kept in culture for at least 1 year, whereas 2D cultures typically deteriorate after 8-12 weeks.
- (3) Maturity, including the formation of dendritic mushroom spines and robust electrophysiological activity.
- (4) Cell type diversity including a more physiological ratio of inhibitory and excitatory neurons (10% GAD67+/NeuN+ neurons in adherent cortical organoids, vs 1% in 2D neural networks), and the emergence of oligodendrocyte lineage cells.

On the other hand, limitations of adherent cortical organoids compared to 2D neural network cultures include:

- (1) Culture times for organoids are much longer than for 2D cultures and the method can therefore be more laborious and more expensive.
- (2) Whole cell patch clamping is not easily feasible in adherent cortical organoids because of the restrictive geometry of 384-well plates.

*(2) It would be important to further benchmark the throughput, ie what is the success rate in filling and successfully growing the organoids in the entire 384 well plate?*

We have addressed this question in the current version of Fig. S1C, in which multiple batches of organoids of all three lines were scored for their success rate. The graph reflects the proportion of properly formed organoids of +/- 400 seeded wells scored at different timepoints, in which each timepoint is a different batch. As mentioned in the response to Reviewer 1, we have also added data on the number of organoids seeded per line in the figure legend.

*(3) For each NPC line an optimal seeding density was estimated based on the proliferation rate of that NPC line and via visual observation after 6 weeks of culture. It*



would be important to delineate this protocol in more robust terms, in order to enable reproducibility with different cell lines and amongst the labs.

As outlined in the response to Reviewer 1, we have clarified the Methods and Discussion sections on seeding density and proliferation rate.

**Reviewer #3 (Recommendations for the authors):**

*Kroeg et al. have introduced a novel method to produce 3D cortical layer formation in hiPSC-derived models, revealing a remarkably consistent topography within compact dimensions. This technique involves seeding frontal cortex-patterned iPSC-derived neural progenitor cells in 384-well plates, triggering the spontaneous assembly of adherent cortical organoids consisting of various neuronal subtypes, astrocytes, and oligodendrocyte lineage cells. Compared to existing brain organoid models, these adherent cortical organoids demonstrate enhanced reproducibility and cell viability during prolonged culture, thereby providing versatile opportunities for high-throughput drug discovery, neurotoxicological screening, and the investigation of brain disorder pathophysiology. This is an important and timely issue that needs to be addressed to improve the current brain organoid systems. While the authors have provided significant data supporting this claim, several aspects necessitate further characterization and clarification. Particularly, highlighting the consistency of differentiation across different cell lines and standardizing functional outputs are crucial elements to emphasize the future broad potential of this new organoid system for large-scale pharmacological screening.*

*(1) Considering the emergence of astrocyte markers (GFAP, S100b) and upper layer neuron marker (CUX1) around Day 60, the overall differentiation speed is significantly faster compared to other forebrain organoid protocols. Are these accelerated sequences of neurodevelopment consistent across different hiPSC lines?*

As shown in Fig. S5, astrocytes are present around Day 60 for all three lines. For comparison with other organoid protocols, an important consideration is that the timeline for these organoids starts at NPC plating, while for other protocols timing often starts from the hiPSC stage. We have clarified the timeline in the graphical abstract in Figure 1A and in the Methods.

*(2) The calcium imaging results in Figure 4G were recorded at a single time point, Day 61, a relatively early time window compared to other forebrain organoid protocols (more than 100 days, PMID: 31257131; PMID: 36120104). Are the neurons in adherent cortical organoids functionally mature enough around Day 61? How consistent is this functional activity across different cell lines and independent differentiation batches?*

As discussed above in Point 1, it is important to consider that the specified timeline starts from NPC plating. In analogy to 2D neural networks, robust neuronal activity can be observed after ~8 weeks in culture. In addition, we have now added calcium imaging data for an additional batch of organoids at Day 100 in Figure 5, which exhibit comparable levels of neuronal activity as observed on Day 61.

*(3) Along the same line, Various cell types, such as oligodendrocytes and astrocytes, are believed to influence neuronal maturation. Therefore, longitudinal studies until the late stage are necessary to observe changes in electrophysiological activity based on the degree of neuronal maturation (at least two more later time points, such as 100 days and 150 days).*

As described in the previous points, we have now included a Day 100 time point in the calcium imaging data, in addition to the recordings at Day 61 (Figure 5C-E).

*(4) The authors assert that heterogeneity among organoids has been diminished using the human adherent cortical organoids protocol. However, there is inadequate quantitative data to prove the consistency of neuronal activities between different wells. Therefore, experiments quantifying the degree of heterogeneity between organoids, such as through methods like calcium imaging, are necessary to determine if neuron activity occurs consistently across each organoid well.*

We agree with the review and have added several quantitative experiments: a) we've added another timepoint to the calcium imaging data in Figure 5 from an independent batch of 3 adherent cortical organoids, which demonstrates continued robust activity at day 100, as well as day 61; b) we added synapse quantification in Figure 4, and c) interneuron quantification in Figure 2. We are currently also pursuing high throughput measures of activity to assess the longitudinal activity of ACOs in a larger number of wells. This way we can more definitively quantify the time-dependent variance in organoid activity.

*(5) Is this platform applicable to other functional measurements for neuronal activity, such as the MEA system? When observing the morphology of neurons formed in organoids, they appear to extend axons and dendrites in a consistent direction, suggesting a radial structure that demonstrates high reproducibility across wells. A culture system where neurons are arranged with such consistency in directionality could be highly beneficial for experiments utilizing the MEA system to assess parameters such as the speed of electrical activity transmission and stimulus-response. Therefore, there seems to be a need for a more detailed explanation of the utility of the structural characteristics of the culture system.*

The ACO platform is indeed suitable for MEA recordings. We are in the process of engineering the required geometry using HD-MEA systems through specialized inserts to generate ACOs on MEA systems.

*(6) In Figure 2E-I, authors suggest morphological diversity of GFAP+/S100b+ astrocyte, but the imaging data presented in Figure F-I is only based on GFAP immunoreactivity.*

*Since GFAP is also expressed in radial glial cells at this stage (Figure 2I), many fibrous astrocytes and interlaminar astrocytes are likely radial glial neural progenitor cells instead of astrocytes. It appears necessary to perform additional staining using astrocyte markers such as S100B or outer radial glia markers such as HOPX to demonstrate that the figure depicts subtype-specific morphologies of astrocytes.*

In Figure 2M, we stained for GFAP and PAX6 to mark radial glia that look different than the astrocyte morphologies we describe in Figure 2J-L. We see a large overlap in GFAP and S100B staining in Figure 2I, in which most GFAP+ cells are double positive for S100B (yellow) that is more consistent with astrocyte maturation than radial glia. Furthermore, we have not seen PAX6 staining outside the dense edges of the center of the ACO.

*(7) In Figure 4D, the axon appears to exhibit directionality. Additional explanation regarding the organization of the axon is necessary. Further research utilizing sparse staining to examine the morphology of single neurons seems warranted.*

The polarized directionality of the axons is something we indeed have also noticed. We are looking into options to further investigate this intriguing property of the ACOs.

*(8) Figure 1E-F only showed cell viability in the early stages around Day 40-50. To demonstrate the superior long-term viability of ACO culture, it appears necessary to illustrate the ratio of dead cells to live cells over the course of a time course.*

Figure S1B shows LIVE/DEAD staining for ACOs of all three lines, revealing minimal DEAD staining at Day 56. A longitudinal time course experiment was not performed, however the line- and batch-specific quantifications over developmental timepoints in Figure S1C provide an indication of the robust long-term viability of the ACOs.

<https://doi.org/10.7554/eLife.98340.2.sa0>